SlideGen-Q: Benchmark-Validated Technical Slide Evaluation for Content-First Generation

Anonymous Author

Anonymous Institution
anonymous@example.com

Abstract. Automatic slide generation has advanced, but its evaluation remains poorly standardized. Existing protocols rely on aesthetic scores, human preference, or opaque vision-language-model judgments, offering limited criterion-level diagnosis and no unified evidence from rendered appearance and executable artifacts. Technical quality is difficult to compare, reproduce, and audit. We address this gap with a 4,104-image benchmark covering five defects: background color, text color, font size, element overlap, and text overflow. Its clean and corrupted slides are manually constructed, visually verified, and programmatically audited; group-disjoint train–test partitioning prevents variants from crossing splits. We develop a hybrid post-generation evaluator combining frozen XML and geometry rules with a shared multi-label Vision Transformer (ViT). The shared ViT-Base achieves 0.764 test macro F1, compared with 0.424 for a frozen-backbone linear probe and 0.701 for five independent binary ViTs, while requiring one forward pass. We present SlideGen-Q, a content-first agentic pipeline that plans the deck narrative before retrieving typed layout functions and producing editable XML. Evaluation remains observational and never alters generation. Applied without adaptation to 45 outputs from six heterogeneous documents, the frozen ViT accepts 34 slides and identifies overlap as the dominant warning. A live deployment is available at `https://ptitai.org/slidegen`. Together, the benchmark, auditable evaluator, and executable pipeline provide a reproducible foundation for criterion-level technical slide assessment.

Keywords: Slide quality benchmark · Hybrid slide evaluation · Visual defect detection · Vision Transformer · Agentic presentation generation

1 Introduction

Automatic presentation systems can now synthesize complete, editable decks, yet their progress remains difficult to measure consistently. Human preference studies capture communicative judgment but are costly and difficult to reproduce at scale. Learned aesthetics scores summarize visual appeal, while vision-language-model judges can provide broad critiques; neither form of assessment necessarily identifies which technical defect occurred or exposes evidence that another evaluator can audit [10, 15, 7, 8]. A technically useful evaluation protocol must localize failures rather than reduce a slide to an unexplained scalar.

Rendered appearance is only one view of a generated slide. An image can reveal poor contrast, undersized text, overlap, and overflow, but it does not directly expose malformed XML, invalid function arguments, or browser geometry. Conversely, deterministic program checks establish concrete violations but cannot recognize every perceptual failure. Evaluation therefore requires criterion-level visual supervision together with artifact-level evidence. It also requires leakage-resistant data partitioning: clean and corrupted variants derived from the same source slide must not be separated across training and test sets.

This work addresses these requirements with a manually constructed benchmark of 4,104 slide images and a hybrid post-generation evaluator. Five defect labels describe background color, text color, font size, element overlap, and text overflow. Every rendering is visually checked against its intended labels, an executable audit verifies files and partition integrity, and group-disjoint splitting isolates related slide families. The learned branch uses one shared multi-label ViT, while frozen rules retain explicit XML and geometry evidence. Importantly, only the visual branch is benchmark-validated by the PNG labels; the deterministic branch remains separately traceable rather than being credited with the ViT's measured accuracy.

We study deployment through SlideGen-Q, a content-first agentic generator that determines source structure, deck narrative, and slide intent before retrieving typed layout functions and synthesizing editable XML. This ordering reflects evidence that explicit outline and content planning improve coverage, coherence, and generation success [1, 18]. The completed XML and rendering are immutable inputs to the evaluator: evaluation neither retries generation nor changes the artifact. Thus, the generation system supplies a realistic deployment setting, while the benchmark provides controlled evidence for technical visual-defect detection.

The contributions of this work are:

- We establish a 4,104-image, five-criterion technical slide benchmark whose manually verified labels, executable audit, and group-disjoint train–test protocol support reproducible visual-defect evaluation.
- We develop a hybrid evaluator combining frozen XML/geometry evidence with five learned defect scores. A fine-tuned shared ViT outperforms frozen-backbone and independent-ViT baselines.
- We introduce a content-first agentic generation system with late-bound function retrieval and editable XML, enforcing a one-way boundary that prevents evaluator feedback from altering generation.
- We characterize evaluator transfer on the complete 45-slide output set from six source documents, reporting slide- and deck-level responses together with representative warnings and false-positive failure modes.

The current benchmark directly validates defect detection; it does not substitute for a future human study of end-to-end content quality, coherence, and communicative effectiveness.

2 Related Work

2.1 Visual Quality, Learned Evaluation, and Benchmarks

Aesthetics models learn ratings, composition, spatial structure, and color/style preferences [10, 15, 2, 9, 13, 6]. Slide verification additionally requires legibility and geometric validity, which image-only aesthetic assessment cannot fully establish from hidden program state. ViTs provide global patch interactions [3] for criterion prediction; slide benchmarks and vision-guided refinement motivate rendered assessment [7, 8, 11]. Together, these strands leave open criterion-level, auditable technical assessment that connects rendered evidence with executable slide artifacts.

2.2 Content Planning and Visual Realization

Early systems extracted salient text and figures into predefined layouts [12, 14]. Other work exploited document structure [16], while DOC2PPT jointly modeled summarization, retrieval, paraphrasing, and layout prediction [4]. More recent pipelines make planning explicit: DocPres builds a document overview, outline, source mapping, and context-aware slide text [1], while PPTAgent infers functional types and content schemas before editing references [18]. These systems establish that content selection, narrative, and arrangement are coupled, although template- and reference-conditioned realization can restrict variation. AutoPresent learns editable instruction-to-code generation and refinement for single slides [5]; PreGenie coordinates multimodal summarization, code generation, and rendered-page review [17]. Our pipeline provides executable deployment outputs, whereas the principal empirical contribution concerns their criterion-level post-generation assessment.

3 Deployment System: Content-First Presentation Synthesis

Figure 1 follows the causal order of generation. Document hierarchy or agent research first yields source-grounded slide units; deck planning then assigns rhetorical roles, messages, and design intents. Only afterward does retrieval expose typed layout functions for XML synthesis and rendering. The pipeline terminates with an editable XML program, a rendered slide, and deterministic execution metadata; evaluation is performed separately in Section 4.

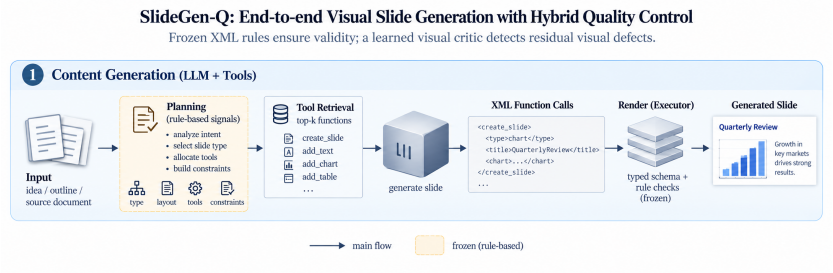


Fig. 1. Content-first SlideGen-Q generation workflow: Input → Planning → Tool Retrieval → Generate Slide → XML Function Call → Render (Executor) → Generated Slide. The post-generation evaluator is intentionally outside this flow.

The *Input* stage preserves or restores document hierarchy and groups evidence into slide-sized content units. In topic-driven mode, a tool-calling agent performs up to three iterations before emitting the same representation. Both input paths determine ideas and ordering without selecting visual placeholders.

For each unit, *Planning* assigns a rhetorical role and content-specific design intent. *Tool Retrieval* then selects top-k typed functions by relevance and diversity, with a temporal penalty discouraging immediate reuse. *Generate Slide* receives the content unit, its plan, neighboring context, prior-slide style, and retrieved signatures; only style attributes propagate across slides, preventing text leakage.

The resulting *XML Function Call* instantiates text, geometry, media, and charts. *Render (Executor)* validates the tool name, typed arguments, legal values, and XML root before producing the editable XML and *Generated Slide*. These outputs and their execution metadata are passed forward as a fixed *Generated Candidate*; no evaluator output is returned to Planning, Tool Retrieval, or Generate Slide.

4 Hybrid Slide Evaluation

4.1 Task Formulation and Scope

For each slide-sized content unit, Planning produces a rhetorical intent, Tool Retrieval returns the top-k functions, Generate Slide emits a typed call, and the executor validates it before returning editable XML and a rasterized slide. Together, the call, XML, and image form a fixed Generated Candidate that is evaluated without feedback or retry.

We evaluate five potentially co-occurring binary defects, denoted by $\mathcal{C}$: background color (BG), text color (TC), font size (FS), overlap (OV), and text overflow (OF). A positive label denotes the presence of a defect.

The *Deterministic Rule Branch* inspects the function call, XML, and rendered slide, while the *Learned Visual Branch* estimates five defect probabilities from the image. The Evaluation Report retains both sources of evidence. Its

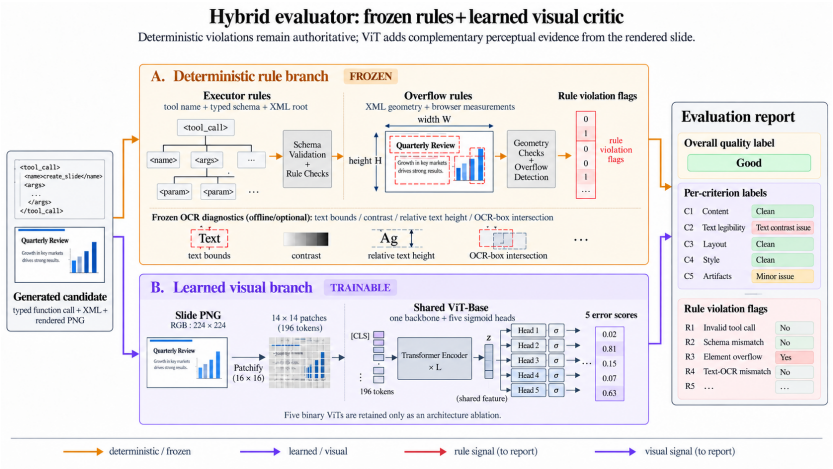


Fig. 2. Post-generation hybrid evaluator. The fixed Generated Candidate is inspected by deterministic and learned visual branches. The Evaluation Report preserves rule violations and five defect probabilities without feeding a signal back to generation.

C1–C5 groups in Fig. 2 illustrate report organization; the benchmark outputs and all experiments use the formal criterion set $\mathcal{C}$. Only the ViT parameters are learned; generation, rendering, rules, OCR, and thresholds remain fixed. Consequently, the report concerns the five technical defects in $\mathcal{C}$ rather than semantic fidelity, communicative effectiveness, or aesthetic preference.

4.2 Frozen Deterministic Branch

Executor rules validate operators, typed arguments, well-formed XML, and execution. Browser measurements expose element overflow and geometric violations that may be invisible in source text. Their positive flags remain authoritative in the technical report because they correspond to directly testable conditions. Frozen OCR/CV diagnostics estimate text bounds, contrast, relative text height, and box intersections from the raster image; because OCR localization is uncertain, these outputs are reported as optional evidence and studied separately in the fusion ablation.

4.3 Shared Multi-Label Visual Defect Detector

Shared ViT-Base encodes each normalized slide once and uses five sigmoid heads to estimate one defect probability per criterion. Independent heads permit defects to co-occur. Training uses class-weighted binary cross-entropy; for criterion i, positive examples receive weight $N_i^- / \max(N_i^+, 1)$, where N_i^+ and N_i^- are its positive and negative training counts. Only the shared encoder and heads are optimized. The five-binary baseline instead trains an independent ViT for each criterion, allowing RQ1 to isolate the effect of feature sharing.

4.4 Evidence Aggregation and Reporting

The Rule Branch returns hard violation flags, while the Visual Branch marks criterion i when its probability reaches threshold τ_i. A slide is reported as technically defective if any rule or learned criterion fires. The report retains every rule flag, learned decision, and raw probability, so a warning remains traceable; no warning means only that no implemented check fired. Frozen OCR/CV warnings are OR-fused with ViT decisions only in the offline fusion experiment.

5 Slide Quality Benchmark

The benchmark contains 4,104 clean, single-error, and dual-error slides. Five contributors manually introduced controlled corruptions and checked every rendering against the five labels in $\mathcal{C}$. These labels measure technical defects rather than factual correctness or narrative quality.

Group-stratified partitioning keeps all variants of one contributor/source-slide family together, yielding a 3,694-image training pool and a locked 410-image test set. Figure 3 reports their composition and criterion support; dual-error slides contribute to two label counts.



Fig. 3. Training-pool and locked-test composition with criterion-level error support. Similar clean, single-error, and dual-error proportions are retained across splits; overlap is the most frequent defect.

An executable audit validates all paths and labels, decodes every image, checks distributions and SHA-256 hashes, and confirms zero path, byte, or slide-family overlap across partitions. Controlled corruptions provide reproducible labels but may not capture the full variation of naturally occurring defects.

6 Experimental Setup

RQ1 compares frozen, shared, and independent ViTs; RQ2 tests OCR/CV fusion; RQ3 measures generated-output transfer; and RQ4 ablates Planning and Tool Retrieval. Table 1 lists the checkpoints.

Table 1. Model and checkpoint configuration used for test evaluation.

Configuration	Backbone/component	Criteria/output	Epoch(s)	Batch
Frozen linear probe	ViT-Base/16-224, IN21K; frozen	All five; linear 5-logit head	20	128
Shared ViT	ViT-Base/16-224, IN21K	All five; one 5-logit head	22	64
Binary color	ViT-Small/16-224	BG, TC; two models	14, 7	128
Binary structural	ViT-Base/16-224, IN21K	FS, OV, OF; three models	8, 15, 25	64
OCR/CV diagnostics	Frozen PaddleOCR and heuristics	BG, FS, OV, OF	–	–

ViT-Base configurations use ImageNet-21K pretraining; color binaries use ViT-Small. Inputs are resized to 224×224, ImageNet-normalized, and augmented by rotation, flipping, and mild color jitter. Fine-tuning uses AdamW (5×10^{-5} learning rate, 5×10^{-3} weight decay), cosine decay, 5% warm-up, mixed precision, seed 45, and early stopping (patience 5; maximum 30 epochs). Of 3,694 training-pool images, 3,324 optimize parameters and 370 select checkpoints. The frozen probe trains only 3,845 head parameters and selects 10^{-3} from $\{10^{-4}, 5 \times 10^{-4}, 10^{-3}\}$ by internal macro F1. Experiments run on an NVIDIA RTX 5880 Ada; rules, OCR, and heuristic thresholds remain fixed.

All primary results use threshold 0.5. Criterion-specific thresholds are selected only on the internal model-selection split for sensitivity analysis; test labels are never used for threshold selection. Metrics include criterion-level precision, recall, F1, accuracy, AUROC, and AUPRC, together with aggregate F1, exact match, Hamming loss, slide-level detection, and inference latency. They measure technical defects, not semantic or communicative quality.

7 Results

7.1 ViT Benchmark Results

Table 2 compares every criterion at the fixed 0.5 threshold. The shared ViT outperforms the selected binary checkpoint on all five F1 scores, although the size of the gain varies substantially.

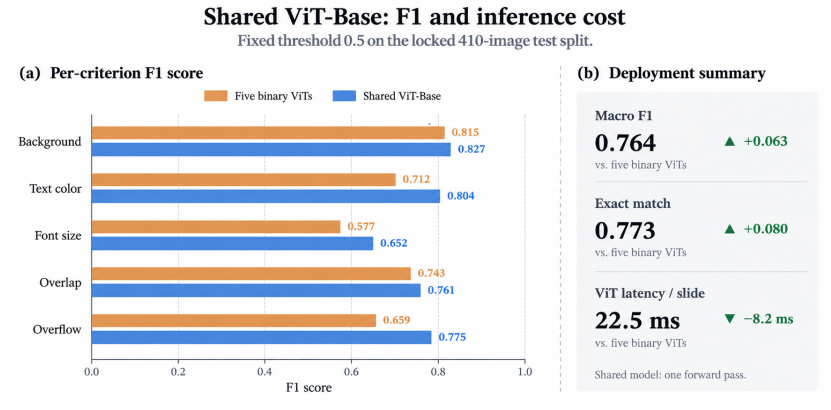


Fig. 4. Fixed-threshold comparison of the two fine-tuned architectures. The shared ViT improves all five criterion F1 scores, macro F1, and exact match while replacing five sequential forward passes with one.

Table 2. Criterion-level test performance for the shared and independent ViT configurations.

Model	Criterion	Prec.	Recall	F1	Acc.	AUROC	AUPRC
Shared ViT	Background color	0.843	0.811	0.827	0.956	0.980	0.915
	Text color	0.765	0.848	0.804	0.954	0.966	0.896
	Font size	0.732	0.588	0.652	0.922	0.885	0.717
	Overlap	0.736	0.788	0.761	0.898	0.918	0.811
	Overflow	0.816	0.738	0.775	0.956	0.924	0.824
Binary ViTs	Background color	0.800	0.830	0.815	0.951	0.974	0.895
	Text color	0.638	0.804	0.712	0.927	0.960	0.774
	Font size	0.533	0.627	0.577	0.885	0.905	0.615
	Overlap	0.756	0.729	0.743	0.895	0.889	0.800
	Overflow	0.651	0.667	0.659	0.929	0.922	0.778

The shared model improves every criterion. The largest F1 gains are 0.116 for overflow, 0.093 for text color, and 0.076 for font size. Font size remains the weakest criterion (0.652 F1), consistent with loss of small-text detail at 224×224. The frozen probe reaches only 0.424 macro F1 and 0.276 exact match; fine-tuning raises these to 0.764 and 0.773, respectively. Figure 4 summarizes the fine-tuned comparison.

Despite similar macro AUROC (0.935 versus 0.930), the shared model exceeds the binary ensemble by 0.063 macro F1 and 0.047 slide-level F1, while reducing sequential inference time from 30.69 to 22.46 ms per slide (26.8%). Validation-selected thresholds do not improve either fine-tuned variant, supporting 0.5 as the primary operating point.

7.2 Frozen Baselines and OCR/CV Fusion

The frozen OCR/CV baseline covers four criteria but has no text-color rule. Its covered-criterion F1 is 0.414, while slide-level recall and F1 are 0.917 and 0.689, indicating a conservative, high-recall warning system.

OR fusion (Table 3) raises shared-ViT macro recall from 0.755 to 0.878 and slide-level recall from 0.865 to 0.984, but lowers macro F1 from 0.764 to 0.564 and exact match from 0.773 to 0.268. It is therefore useful for recall-oriented screening, not as the primary benchmark score. Since the benchmark contains PNGs only, this experiment excludes the XML executor and schema rules in Fig. 2.

Table 3. Aggregate test results and OCR/CV–ViT OR fusion at threshold 0.5. The frozen probe shares the one-pass inference graph of Shared ViT; latency is milliseconds per slide.

Configuration	Prec.	Recall	F1	Exact	Hamming	Lat.
Frozen ViT probe	0.310	0.685	0.424	0.276	0.257	–
OCR/CV	0.270	0.525	0.331	0.256	0.262	2460.1
Shared ViT	0.778	0.755	**0.764**	**0.773**	**0.063**	**22.5**
OCR/CV + shared	0.452	**0.878**	0.564	0.268	0.238	2482.6
Binary ViTs	0.676	0.732	0.701	0.693	0.082	30.7
OCR/CV + binary	0.409	0.866	0.531	0.254	0.249	2490.8

7.3 Full-System Quantitative Evaluation

We apply the epoch-22 shared ViT, without adaptation, to all 45 generated PNGs from six heterogeneous documents; no output is filtered. Because the archive lacks XML/browser traces, this experiment evaluates only the learned visual branch.

At threshold 0.5, a generated slide passes when none of the five ViT probabilities reaches the threshold. We report the fraction of passing slides and $\bar{p}$, the mean of all five probabilities across all slides, aggregated by slide and deck. These are evaluator-estimated responses because the generated slides lack human defect labels; a pass does not certify semantic or overall visual quality.

Table 4 reports 34/45 accepted slides (75.6%). Across six decks, pass rate is $75.5\% \pm 9.2\%$ and mean defect probability is $9.9\% \pm 2.9\%$ (mean $\pm$ sample standard deviation). Batched inference takes 0.98 s, or 21.72 ms per slide, excluding loading and rendering.

Table 4. Frozen shared-ViT assessment of all generated outputs at threshold 0.5. $\bar{p}$ averages five defect probabilities.

ID	Source topic	Slides	Pass (%)	Defect (%)	$\bar{p}$	Labels/slide
D1	Urban culture	8	87.5	12.5	0.076	0.250
D2	Sports biography	7	71.4	28.6	0.118	0.429
D3	Business planning	12	66.7	33.3	0.102	0.333
D4	Historical narrative	3	66.7	33.3	0.134	0.333
D5	Artificial intelligence	8	75.0	25.0	0.107	0.375
D6	Presentation automation	7	85.7	14.3	0.055	0.143
All generated slides		45	**75.6**	24.4	0.096	0.311

Eleven slides receive 14 warnings because three have two labels (Table 5). Overlap dominates with 8 warnings and the highest mean probability (0.189); overflow and font size occur 3 and 2 times, text color once, and background color never.

Table 5. Criterion-level warnings on 45 generated slides. Counts and rates are evaluator predictions, not human-adjudicated defect prevalence.

Criterion	Flagged slides	Rate (%)	Mean probability
Background color	0	0.0	0.032
Text color	1	2.2	0.047
Font size	2	4.4	0.118
Overlap	8	17.8	0.189
Text overflow	3	6.7	0.093

7.4 Full-System Qualitative Evaluation

Figure 5 presents all 45 outputs from six heterogeneous documents without post-generation filtering. The decks retain within-document visual continuity while varying hierarchy, density, image placement, and composition with slide intent rather than collapsing to one fixed template. The complete plate exposes both successful and weaker outputs, supporting qualitative inspection of content–layout correspondence and structural diversity. Factual fidelity, narrative preference, and audience effectiveness remain outside this visual analysis and require blinded human evaluation.

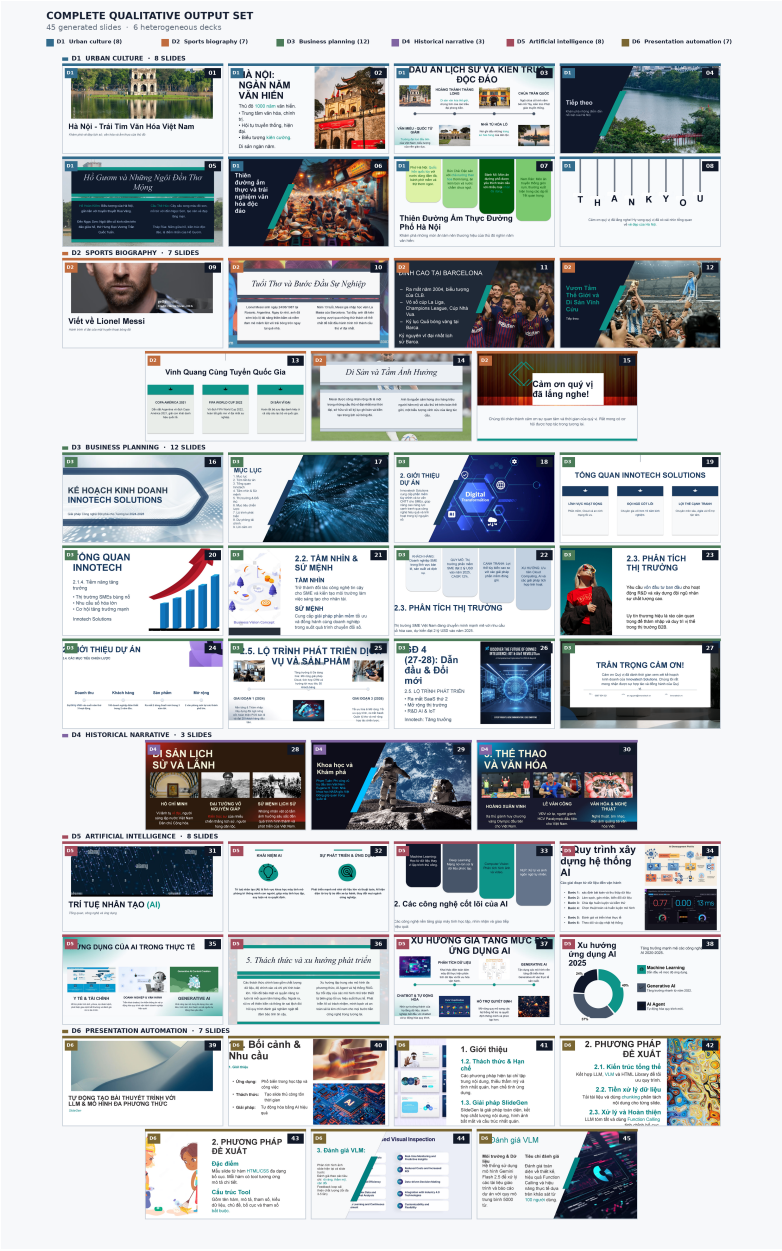


Fig. 5. Complete output set comprising all 45 slides generated for six heterogeneous source documents, shown without post-generation filtering. Deck headers and badges D1–D6 identify the source document of every slide; badges 01–45 provide stable slide references for visual inspection. This plate is descriptive and does not replace criterion-level labels or blinded human assessment.

8 Error Analysis

Figure 6 contrasts plausible, ambiguous, and implausible warnings. Dense text in (a), small body copy in (c), and compact timeline labels in (d) indicate legibility risks, although (a) and (d) are assigned to overlap rather than density. Conversely, the separated title layout in (b) and chart in (f) are likely overlap false positives; their primitives may resemble intersections in corrupted training slides. Panel (e) is ambiguous because its narrow text column suggests density without visible overflow. Thus, ViT flags support screening but do not prove the reported criterion, and the 75.6% pass rate remains evaluator-estimated rather than human-adjudicated.

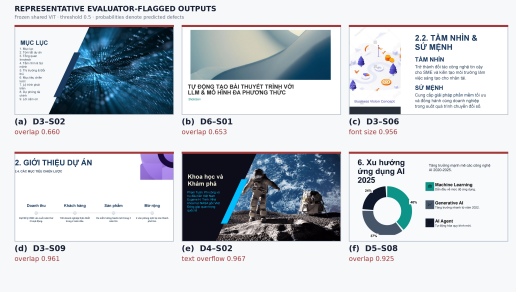


Fig. 6. Representative ViT warnings: plausible concerns in (a), (c), and (d), an ambiguous case in (e), and likely false positives in (b) and (f).

9 Ablation Study

9.1 Quantitative Evaluation

We run an exploratory paired ablation on 25 aligned chunks from three documents. The full mode uses Planning, design preparation, and top-5 Tool Retrieval; *No planner* queries retrieval with the raw content unit, *No retriever* exposes the complete function category, and *Without both* combines these removals. Rules remain frozen and ViT evaluation is post-generation. Each condition is sampled once, so the study does not establish population-level superiority.

Hybrid acceptance requires both a negative hard-overflow check and no ViT warning among the five criteria. All 100 finalized artifacts satisfy tool-call and XML validity checks. Lexical source recall is the fraction of case-folded alphanumeric source tokens retained in generated XML; it does not measure paraphrase or semantic fidelity.

Table 6. Paired ablation on three documents and 25 slides per mode. Hybrid combines the hard-overflow and ViT decisions; Hard OF counts overflow warnings; Lex. and Num. denote source-token and source-number recall.

Mode	Hybrid (%)	ViT (%)	Hard OF	$\bar{p}$	Lex.	Num.
Full	64.0	64.0	**0**	0.129	**0.567**	0.737
No planner	**80.0**	**80.0**	**0**	**0.080**	0.552	0.694
No retriever	76.0	**80.0**	2	0.098	0.455	**0.822**
Without both	72.0	**80.0**	2	0.083	0.453	0.727

Table 6 reveals a trade-off. No planner has the highest hybrid pass rate (80.0%), while full reaches 64.0%. Hard overflow further rejects one no-retriever slide and two without-both slides, yielding 76.0% and 72.0% hybrid pass. Conversely, retrieval-enabled modes retain more source text (0.567 and 0.552 versus 0.455 and 0.453); removing retrieval from full reduces recall by 0.112. XML complexity correlates only weakly with defect probability ($|\rho| \leq 0.191$). Thus, RQ4 shows that fewer warnings do not imply better content retention.

9.2 Qualitative Evaluation

Figure 7 visualizes the same trade-off. In the business example, retrieval preserves more source content, whereas joint removal yields a nearly empty slide. For the AI cover, source recall is unchanged but the no-retriever composition is malformed and flagged. In the presentation-method example, structurally different layouts receive different warnings despite aligned content. The cases demonstrate useful anomaly screening but also criterion uncertainty; they are not human preference labels.



Fig. 7. Source-aligned qualitative ablation. Columns vary Planning and Tool Retrieval; rows hold content fixed. Bands report hybrid acceptance, ViT warnings, hard overflow, and lexical recall.

10 Analysis and Discussion

RQ1 separates representation adaptation from architecture: fine-tuning raises macro F1 from 0.424 to 0.764, and the shared model exceeds five binary ViTs by 0.063 macro F1 and 0.047 slide-level F1 while cutting latency by 26.8%. RQ2 shows that OCR/CV OR fusion is useful only when recall is prioritized; it reaches 0.984 slide-level recall but loses 0.200 macro F1 and adds 2.46 s per image. XML/browser invariants remain outside the PNG benchmark.

For RQ3, the 75.6% pass rate characterizes evaluator transfer, not accuracy on generated slides because no human labels are available. Overlap accounts for 8 of 14 warnings, including plausible density issues and false positives around diagonals, timelines, and charts; warnings therefore support screening rather than geometric proof. RQ4 likewise reveals a trade-off: removing Planning improves technical acceptance, whereas Tool Retrieval preserves about 0.11 more source content. Technical warnings must therefore remain separate from narrative-quality evidence.

The evidence remains bounded: the group-disjoint split does not establish generalization across domains, languages, templates, or natural defects; deployment evaluation covers six unlabeled decks; and ablation uses three documents with one generation per condition. At 224-by-224 resolution, small text may vanish while charts and oblique partitions may resemble corruptions [2, 9]. Latency excludes generation, rendering, and loading, and the five-label taxonomy does not measure content preservation, coherence, or audience preference.

11 Conclusion

We establish a manually verified and programmatically audited benchmark for five criterion-level technical slide defects. On its group-disjoint test set, one shared multi-label ViT reaches 0.764 macro F1 versus 0.424 for a frozen-backbone linear probe and 0.701 for five binary ViTs, demonstrating the value of both domain fine-tuning and feature sharing. Frozen XML and geometry evidence complements this benchmark-validated visual branch within a hybrid evaluator whose reports cannot alter generation. SlideGen-Q supplies a content-first, executable deployment setting; on 45 generated outputs, the frozen ViT assigns a 75.6% pass rate and identifies overlap as the dominant warning, while error analysis exposes transfer-related false positives. A paired component study further reveals that fewer technical warnings can coincide with lower literal source retention, reinforcing the need to keep technical and content evidence separate. These findings support reproducible technical screening rather than semantic or audience-level quality and motivate deployment-domain calibration and blinded human evaluation.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

References

1. Bandyopadhyay, S., Maheshwari, H., Natarajan, A., Saxena, A.: Enhancing presentation slide generation by llms with a multi-staged end-to-end approach. arXiv preprint arXiv:2406.06556 (2024)
2. Chen, Q., Zhang, W., Zhou, N., et al.: Adaptive fractional dilated convolution network for image aesthetics assessment. In: CVPR. pp. 14114–14123 (2020)
3. Dosovitskiy, A., Beyer, L., Kolesnikov, A., et al.: An image is worth 16x16 words: Transformers for image recognition at scale. In: International Conference on Learning Representations (2021)
4. Fu, T.J., Wang, W.Y., McDuff, D., Song, Y.: Doc2ppt: Automatic presentation slides generation from scientific documents. In: AAAI. vol. 36, pp. 634–642 (2022)
5. Ge, Y., et al.: Autopresent: Designing structured visuals from scratch. In: CVPR (2025)
6. He, S., Ming, A., Li, Y., et al.: Thinking image color aesthetics assessment: Models, datasets and benchmarks. In: ICCV. pp. 21838–21847 (2023)
7. Huang, Z., Liu, X., Hu, T., Zhang, K., Liu, Y.: Pptbench: Towards holistic evaluation of large language models for powerpoint layout and design understanding. arXiv preprint arXiv:2512.02624 (2025)
8. Kang, H., Bao, E., Goswami, A.: Vlm-slideeval: Evaluating vlms on structured comprehension and perturbation sensitivity in ppt. arXiv preprint arXiv:2510.22045 (2025)
9. Mai, L., Jin, H., Liu, F.: Composition-preserving deep photo aesthetics assessment. In: CVPR. pp. 497–506 (2016)
10. Murray, N., Marchesotti, L., Perronnin, F.: Ava: A large-scale database for aesthetic visual analysis. In: CVPR (2012)
11. Sansford, H., Law, D.H.C., Liu, W., Tripathi, A., Agarwal, N., van den Burg, G.J.J.: Vision-guided iterative refinement for frontend code generation. arXiv preprint arXiv:2604.05839 (2026)
12. Sefid, A., Wu, J., Mitra, P., Giles, C.L.: Automatic slide generation for scientific papers. In: BIRNDL (2019)
13. She, D., Lai, Y.K., Yi, G., Xu, K.: Hierarchical layout-aware graph convolutional network for unified aesthetics assessment. In: CVPR. pp. 8475–8484 (2021)
14. Sravanthi, M., Chowdary, C.R., Kumar, P.S.: Slidesgen: Automatic generation of presentation slides for a technical paper using summarization. In: FLAIRS (2009)
15. Talebi, H., Milanfar, P.: Nima: Neural image assessment. IEEE Transactions on Image Processing **27**(8), 3998–4011 (2018)
16. Ueda, T., et al.: Research on automatic generation of presentation slides from a latex manuscript of a paper. Journal of Japan Society for Fuzzy Theory and Intelligent Informatics **18**(5), 752–760 (2006)
17. Xu, X., Xu, X., Chen, S., Chen, H., Zhang, F., Chen, Y.C.: Pregenie: An agentic framework for high-quality visual presentation generation. In: Findings of the Association for Computational Linguistics: EMNLP 2025. pp. 3045–3063 (2025). https://doi.org/10.18653/v1/2025.findings-emnlp.165
18. Zheng, H., Guan, X., Kong, H., et al.: Pptagent: Generating and evaluating presentations beyond text-to-slides. In: EMNLP (2025)